Hydroponic Primer & Library Map

Guide 4 · Library Front Door

Get Started · Find Your Way · Master Calendar

Contents

RELATED GUIDES

This guide is the front door to the Dragoon Mountains Growing Library:

- **Guide 1 — Peppers in the Dragoons** · soil + hydroponic + staking — the cornerstone fruiting-crop guide, with the full Rocoto protocol
- **Guide 2 — Bato Bucket Hydroponic Systems** · system build, water-treatment protocol, daily operations — the canonical home for the 5-step Cochise bicarbonate scrub
- **Guide 3 — Hydroponic Nutrients** · Masterblend Part 1, locally-sourced DIY Part 2, pH-adjustment recipes Part 3, indoor/LED Part 4
- **Guide 4 — Hydroponic Primer & Library Map** (this guide) · the orientation map, environmental baseline, and master multi-crop calendar
- **Guide 5 — Grapes in the Dragoons** · Cochise County viticulture, VSP trellising, late-pruning frost strategy
- **Guide 6 — Cool-Season Crops** · brassicas + alliums + greens + roots — the October-through-May complement to the warm-season fruiting crops

About this library

What it is · who it is for · the short history

The Dragoon Mountains Growing Library is a six-guide reference set for serious household food and wine growing on the east side of the Dragoons, at roughly 4,500 – 5,500 ft elevation in Cochise County, southeastern Arizona. The guides sit beside each other rather than stacking on top of each other: pick the one that answers your current question, follow its cross-references where they point, and you will not have to read all six to plant a crop.

This is Guide 4, the front door. It is the one to read first if you are new — and the one to come back to when you cannot remember which guide covers a given subject. Everything in this primer is also covered more thoroughly somewhere else in the library; nothing important lives in only one place except the things explicitly named as canonical (the bicarbonate water-treatment protocol in Guide 2 §4, the Masterblend recipes in Guide 3 Parts 1–2, the pH recipes in Guide 3 Part 3).

A short history of the library

The original Dragoon Mountains Growing Guides ran to eight short PDFs lettered A through H — a pepper guide, a Bato bucket build, a hydroponic pepper guide, this primer, a Masterblend guide, a foraged-nutrients guide, a pH-up/pH-down guide, and a grape guide. They were useful but overlapping: the pepper guide had its own water-treatment paragraph, the bucket guide had a slightly different one, the primer had a third, and nobody could remember which to trust. The Masterblend and DIY-nutrient material had drifted apart even though both were trying to deliver the same ppm profile, and pH recipes were split across three different documents.

In spring 2026 the library was consolidated from those eight guides down to six. The pepper and hydroponic-pepper guides were merged into a single Guide 1 with separate soil and hydroponic tracks. The Bato bucket build became Guide 2 and absorbed the canonical water-treatment protocol from every other guide. The Masterblend, DIY-nutrient, pH-replacement, and indoor/LED material were combined into Guide 3 as four numbered Parts. A new Guide 6 was written to cover the entire cool-season palette — brassicas, alliums, greens, roots — which the old library had no real home for. The grape guide became Guide 5. This primer (Guide 4) was rewritten as a navigation document with one major new section: a true multi-crop calendar covering everything that grows here.

> **If you are new here:** read this guide (Guide 4) first, end to end. Then pick **one** of Guide 1 (peppers) or Guide 6 (cool-season) depending on the season you are entering, and read it next. Guide 2 (the Bato build) and Guide 3 (nutrients)

> are reference works — dip into them when you need them, do not try to read them through.

How to use this library — by goal

Find the right guide for what you are actually trying to do

The fastest way through the library is by goal, not by topic. Pick the row that matches what you want to do and follow the cross-reference. Most goals touch two guides; a few touch three.

If you want to…	Start with	Then read
Grow peppers — any species, any system	**Guide 1** entire	Guide 3 Part 1 for the nutrient program (if hydroponic)
Build a Bato bucket system from scratch	**Guide 2** §1–§6	Guide 1 §6 for pepper-specific operation; Guide 3 Part 1 for the recipe
Formulate your own nutrients from local materials	**Guide 3 Part 2**	Guide 3 Part 1 for the target ppm profile to match
Make your own pH-up or pH-down from foraged materials	**Guide 3 Part 3**	Guide 2 §4 for where to use them in the bicarbonate protocol
Plant a vineyard	**Guide 5** entire	Guide 3 Part 3 if you need to acidify irrigation water
Grow lettuce, kale, and brassicas through the winter	**Guide 6** §1, §2, §4	Guide 3 Part 1 §1.4.3 for leafy-greens hydroponic recipes
Plant garlic and overwintering onions	**Guide 6** §3	This guide §9 (master calendar) for exact dates
Treat hard well water for hydroponic use	**Guide 2 §4** (canonical)	Guide 3 Part 3 if you want to make your own acid
Overwinter a Rocoto plant indoors under LEDs	**Guide 1 §10**	Guide 3 Part 4 for the LED program
Set up rainwater catchment for the monsoon	This guide §6 + **Guide 2 §4**	Guide 5 §6 if irrigating vines from catchment
Figure out what to plant in October	This guide §9 (master calendar)	Guide 6 entire
Diagnose a sick plant	This guide §11 (decision tree)	Guide 1 §8 or Guide 3 §1.7 for deficiency tables

LOCAL TIP

The library is built for sideways navigation. Almost every chapter ends with a `relatedguides` callout pointing to the next thing you would naturally want to read. Follow those — they were written for exactly this purpose.

The 6-guide map

A one-paragraph overview of each guide and a topic list

Guide 1 — Peppers in the Dragoons

The cornerstone fruiting-crop guide. Covers soil and hydroponic tracks for every commonly-grown *Capsicum* species, with a dedicated Rocoto section (the species most suited to Dragoon elevation), full staking and pruning, pest and disease management, harvest and drying, and a chapter on overwintering. The two seasonal calendars at the end (soil track and hydroponic track) feed directly into this primer's master calendar.

- Why the Dragoons for peppers · the Chiltepin connection
- Variety reference for all five domesticated *Capsicum* species
- The Rocoto special section (*C. pubescens*)
- Soil track: bed prep, irrigation, nutrition, calendar
- Hydroponic track: system selection, water, EC/pH, nutrients, environmental control, calendar
- Staking, pruning, multi-string trellising per variety
- Pest and disease management
- Harvest, drying, and processing for the Dragoon climate
- Overwintering — which peppers are worth the trouble

Guide 2 — Bato Bucket Hydroponic Systems

The system-build and operations guide. Covers the recommended Bato (Dutch) bucket setup for the Dragoons — bench frame, downspout drain channel, RAINPOINT solar pump, 15-gallon reservoir — plus the canonical 5-step Cochise bicarbonate water-treatment protocol that every other guide references. Crop-agnostic by design: the same build runs peppers, tomatoes, and cool-season crops with parameter changes only.

- Why Bato buckets for Cochise County
- Equipment, components, sourcing
- Bench construction, downspout drilling, medium fill
- **The 5-step Cochise bicarbonate water-treatment protocol** (canonical)
- Plumbing, RAINPOINT solar pump, cycle programming
- Daily and weekly operations; reading runoff; end-of-crop flush
- Adapting the system between crops (peppers → cool-season → tomatoes)
- Materials list and cost breakdown
- System-side troubleshooting

Guide 3 — Hydroponic Nutrients

The nutrient-program reference, organized in four numbered Parts. Part 1 is the purchased Masterblend program with full per-crop feeding charts. Part 2 is the locally-sourced DIY equivalent — oak ash, bat guano tea, caliche calcium, epsomite magnesium. Part 3 holds every pH-adjustment recipe in the library (prickly-pear vinegar, fermented citric acid, oak-ash lye, and the purchased alternatives). Part 4 covers indoor LED growing, VPD, and CO_2 — relevant for winter overwintering and year-round indoor production.

- **Part 1 — Masterblend.** Why 4-18-38; stock-solution preparation; mixing order; per-crop feeding charts (tomato, pepper, leafy greens); EC and pH daily protocol; deficiency diagnosis
- **Part 2 — Locally-sourced DIY.** Target ppm profile; oak ash for potassium; bat guano for nitrogen; caliche for calcium; epsomite for magnesium; micronutrient sourcing; the 8-step DIY workflow
- **Part 3 — pH adjustment.** Six pH-down recipes; four pH-up recipes; comparison tables; universal dosing protocol
- **Part 4 — Indoor environment.** LED basics; PPFD targets; VPD; CO_2; sensor priority list; off-grid solar automation

Guide 4 — Hydroponic Primer & Library Map *(this guide)*

The front door. Use it for orientation, environmental context (climate, water, soils), the master multi-crop calendar, and the troubleshooting decision tree that routes problems to the right specialist guide.

- About the library and how to use it
- Decision tree by goal
- Know your Dragoons — climate, water, wind, sun, soil
- Systems at a glance
- Water — short version (full procedure → Guide 2 §4)
- Nutrients — short version (full recipes → Guide 3 Parts 1–2)
- pH — short version (full recipes → Guide 3 Part 3)
- **Master Dragoons multi-crop calendar** (this guide's unique content)
- Crop selection beyond the named guides
- Troubleshooting decision tree

Guide 5 — Grapes in the Dragoons

Cochise County viticulture, in full. Covers why the Dragoons are a viable wine region (and how they compare to the Sonoita and Willcox AVAs), site selection, variety recommendations, soil preparation and pH management, VSP trellising, irrigation, nutrition, pruning, and the fruit-to-wine handoff. The grape calendar feeds the master calendar in §9 of this primer.

- Why the Dragoons for wine — comparison to Sonoita and Willcox
- Terroir, site selection, frost-pocket avoidance
- Red and white variety recommendations
- Soil preparation and bicarbonate treatment for irrigation
- Planting, vine spacing, VSP trellis
- Irrigation, salt management, nutrition
- Seasonal management calendar
- Spur vs cane pruning; canopy management through monsoon
- Pest, disease, wildlife management
- Harvest by Brix; year-2+ dormancy and fruit-to-wine handoff

Guide 6 — Cool-Season Crops

The fall/winter/spring complement to Guide 1's warm-season work. Covers brassicas (the cool-season anchor), alliums (the long-occupancy crops — garlic, onions, leeks), cool-season greens (spinach, lettuce, mâche, arugula), and root crops (carrots, beets, parsnips, turnips, radishes). Includes a master cool-season planting calendar and the four-year rotation handoff with peppers.

- Common ground for the cool season — Dragoon window, frost behavior, family overview, master cool-season planting calendar
- Brassicas — broccoli, cauliflower, cabbage, kale, brussels sprouts, kohlrabi
- Alliums — garlic, shallots, onions (spring and overwintering), leeks
- Cool-season greens — spinach, lettuce, mâche, arugula, mustards
- Root crops — carrots, beets, turnips, radishes, parsnips
- Pest management for the cool season
- Crop rotation handoff with peppers (4-year cycle)

Know your Dragoons

The environmental baseline — climate, water, wind, sun, soil

The east side of the Dragoon Mountains, at roughly 4,500 – 5,500 ft elevation in Cochise County, is a sky-island grassland-and-oak environment. It is **not low desert** and it is not high desert; it is genuinely cooler than Phoenix, genuinely drier than the Mogollon Rim, and has a sharper diurnal temperature swing than either. This is the canonical home in the library for "what conditions are we actually growing in" — every other guide refers back to this section for environmental context.

Climate

Factor	Condition	What it means for growing
USDA zone	8a (10 – 15 °F average annual minimum)	Long mild winter; supports overwintering brassicas, alliums, garlic in-ground; frost-tender crops need protection or come out by October
Elevation	4,500 – 5,500 ft typical operating range	Cooler summers than Phoenix or Tucson; longer hang time for grapes; 15–20% stronger UV than sea level
Summer high	85 – 95 °F	Manageable; peppers and grapes set fruit reliably; tomatoes need shade in late June
Summer low	55 – 70 °F	Cool nights are an asset — fruit-set is reliable; flavor compounds and capsaicin concentrate
Winter high	55 – 67 °F	Open-field harvest of cold-tolerant crops all winter
Winter low	22 – 32 °F most nights; 3–6 nights below 25 °F in deep January	Hard freezes are short; row cover handles the worst nights
Diurnal swing	30 – 40 °F daily, year-round	Concentrates flavor; stresses unprotected hydroponic reservoirs

Factor	Condition	What it means for growing
Annual rainfall	15 – 22”	About 60% falls July–September during the monsoon
Monsoon	Mid-July to mid-September	8–12” of soft, low-TDS rainwater; 50–70% humidity; afternoon storms
First frost	Late October to early November	Plan pepper turnover for mid-October
Last frost	Mid-March to early April	Spring brassicas race against May heat after last frost
Frost-free days	~180 – 220 at typical elevation	Long enough for any pepper species; long enough for two grape vintages of cane growth
Sun hours	2,800+ hours/year	More than any commercial European wine region; supports any sun-loving crop

> **The Dragoons’ best growing window is fall — September through November.** Cool nights, warm days, clear skies, no monsoon, no frost. Plan your biggest harvests to mature in this window: peak pepper harvest, grape harvest, fall brassicas establishing, garlic going in the ground. This single fact organizes most of the planting calendar in §9.

Water

The valleys below the Dragoons — Sulphur Springs, San Pedro, Whetstone — pull groundwater from carbonate-bearing aquifers. The result is stubbornly hard, alkaline, bicarbonate-rich well water that is **the single most consequential variable in hydroponic growing here**, and a meaningful one even for soil growing on the alkaline native dirt.

Source	TDS	pH	Treatment needed
Monsoon rainwater (collected)	< 20 ppm	6.0 – 6.5	Minimal — just pH fine-tune
RO-filtered well water	< 50 ppm	6.5 – 7.0	Light pH adjustment

Source	TDS	pH	Treatment needed
Untreated well water	200 – 500 ppm (some wells > 600)	7.5 – 8.5	**Full 5-step bicarbonate protocol**
Surface / stock-tank water	Highly variable	Highly variable	Test before use; may contain pathogens

> **LOCAL TIP**
>
> The well at our place runs about 330 ppm TDS at pH 8.1 — workable with the full protocol but never trivial. If your well measures over 500 ppm consider a small under-counter RO unit purely for hydroponic use; over 600 ppm it is essentially required. **Always meter your own well** before designing a hydroponic system — the variance between wells in Cochise County is large enough that you cannot rely on a neighbor's number.

A 500 sq ft covered patio roof collects 3,000+ gallons of low-TDS rainwater per monsoon season — that is the entire summer reservoir requirement for a 20-bucket Bato system. Catchment is the single most-leveraged piece of infrastructure on the property. Full procedure for treating well water lives in **Guide 2 §4**.

Wind, sun, and air

Wind matters in the Dragoons but is rarely catastrophic at the typical garden elevation. Sustained prevailing winds run 5–15 mph from the southwest; spring (March–May) brings the windiest stretch, with occasional gusts above 35 mph that can shred unsupported pepper plants and tear improperly-secured shade cloth. Monsoon storms bring short, intense gusts from the east — train trellises and stakes accordingly.

UV intensity is roughly 15–20% higher than sea level. This deepens flavor and pigment in fruit but also bleaches leaves, degrades plastic, and overheats unshaded reservoirs. **30–40% shade cloth from mid-June through early September** is essential for most crops; black/yellow UV-stabilized reservoir tubs are essential year-round.

Soil character

Native soils on the east side of the Dragoons are mostly sandy loams developed over granite or alluvial caliche. The defining features for a grower:

Soil property	Typical reading	Implication
pH	7.5 – 8.5	Iron and phosphorus locked up; sulfur amendment needed for acid-loving crops; many cool-season crops are happy as-is
Organic matter	1 – 2%	Native soils are low — needs 2–4” of finished compost worked in for any annual vegetable bed
Caliche layer	12 – 36” below surface, variable	Restricts deep root growth; break through at planting for grapes and Rocoto
Drainage	Generally excellent	Sandy texture and slope drain well; rare to have wet-feet problems outside monsoon
Salt accumulation	Mild but real	Monsoon naturally flushes salts; well-water irrigation can accumulate them; deep-water periodically
Native fertility	Calcium-rich, potassium-moderate, low N and P	Caliche supplies calcium freely; nitrogen and phosphorus must be added
Iron in soil	Present but locked up by high pH	Foliar iron useful early-season; oak-tannin chelation works (see Guide 3 Part 2)

WARNING

Caliche is not negotiable. When you dig a hole for a grape vine or a long-lived Rocoto and hit a layer of cement-like calcium carbonate, you must either break through it (rock bar, pickaxe, sometimes a rented breaker hammer) or move the planting hole. Roots cannot penetrate intact caliche. Half-hearted hole prep is the single biggest cause of slow-failing perennial plantings in the Dragoons.

Systems at a glance

The eight common growing approaches — what they are good for, where to read

There is no single “right system” for the Dragoons. The right choice depends on the crop, the water source, the available infrastructure, and how much daily attention you want to spend on the garden. This section is the comparison table — see the cross-referenced guide for the actual build.

System	Best for	Dragoon strength	Dragoon weakness	Read
In-ground soil	Long-lived perennials (grapes, Chiltepin, garlic); deep-rooted crops (parsnip, carrot)	Native soil is well-drained; calcium-rich from caliche	Alkaline pH locks up iron; needs heavy compost amendment	Guide 1 §5 (peppers); Guide 5 (grapes); Guide 6 (cool-season)
Raised beds (soil)	Brassicas, alliums, greens, roots — anything annual	Bypasses the worst caliche pockets; warmer in spring; easier to amend	Dries out faster in pre-monsoon June; needs mulch	Guide 6 entire
Containers (soil/coco)	Movable plantings (Rocoto, herbs); patio growing; overwintering	Move indoors against frost; control mix exactly	Heat-stress through summer afternoons; constant watering	Guide 1 §5, §10
Bato (Dutch) buckets	Large fruiting crops (peppers, tomatoes, cucumbers, eggplant)	Recirculates expensive treated water; elevated above wildlife; modular	Reservoir-temperature control needed in pre-monsoon heat	**Guide 2 entire** — the canonical build
NFT (Nutrient Film)	Leafy greens, herbs, strawberries — small-root crops only	Highly water-efficient	Cannot run on poorly-treated well water (precipitation); needs constant power	Brief in Guide 6 §4.8

System	Best for	Dragoon strength	Dragoon weakness	Read
DWC (Deep Water Culture)	Tomatoes, peppers, cucumbers; herb starts	Simple, high-yield, easy temperature management with a buried tank	Air pump required; reservoir-temperature critical	Brief in Guide 1 §6.1
Kratky (passive)	Lettuce, basil, single-harvest greens	No pump, no electricity — ideal off-grid; very quick to set up	Single-crop cycle (no top-off mid-grow); not for fruiting crops	Brief in Guide 6 §4.8
Ebb-and-flow / flood-and-drain	Mid-size fruiting crops; trays of seedlings	Flexible; good for nursery propagation	More plumbing complexity than Bato; more pump cycling	Not covered in depth — apply Guide 2 water and Guide 3 nutrients

For the Dragoons, the **default recommendation** is a Bato bucket bench for warm-season fruiting crops (peppers, tomatoes, cucumbers) and a mix of raised beds + Kratky/NFT for cool-season greens and brassicas. This pairing covers more than 90% of household production and shares one set of nutrient stocks. Start here unless you have a specific reason to do otherwise.

Water — short version

The 5-step protocol in one page · full procedure → Guide 2 §4

Cochise County well water is hard, alkaline, and high in bicarbonates. The bicarbonates are the active problem: they act as a chemical buffer that neutralizes acid as fast as you add it, then pushes pH back up overnight. The fix is to neutralize them **before** adding any nutrients. This procedure is called the bicarbonate scrub or the 5-step Cochise protocol; it is the foundation of every hydroponic system in this library.

The 5 steps (compressed)

Step	Action	What happens
1. Fill	Fill reservoir with source water	Starting pH likely 7.5–8.5; EC 0.3–0.8
2. Acid in	Add acid slowly while stirring (citric stock, phosphoric pH Down, or prickly-pear vinegar)	Water fizzes — CO_2 is releasing
3. Wait	Wait until fizzing **completely** stops, 5–15 minutes	Bicarbonates are now neutralized
4. pH adjust	Fine-tune to 5.8–6.0 (or 6.0–6.5 for brassicas/grapes)	pH is now stable — will not bounce back
5. Nutrients	Add Masterblend → Epsom → calcium nitrate, dissolved one at a time	Nutrients dissolve into stable, pre-treated water

WARNING

Never skip Step 3. Adding nutrients while bicarbonates are still active causes calcium and magnesium to precipitate within hours — visible as cloudiness or white film on the tub. Your nutrients are now locked up and your plants are starving.

Always add calcium nitrate last. This rule appears in every hydroponic guide in this library because violating it causes immediate calcium-sulfate precipitation regardless of how well the water was treated.

RELATED GUIDES

For the full procedure — acid selection (prickly-pear vinegar, citric stock, phosphoric, sulfur burn), base selection (oak-ash lye, KOH, baking soda), water-source ranking, and step-by-step dosing — see **Guide 2 — Bato Bucket Hydroponic Systems, §4**. For the acid and base recipes themselves see **Guide 3 Part 3**.

Nutrients — short version

Masterblend vs DIY in one page · full recipes → Guide 3 Parts 1–2

The same target nutrient profile drives both the purchased Masterblend program (Guide 3 Part 1) and the locally-sourced DIY program (Guide 3 Part 2). They are two paths to the same destination. Pick the path that fits your priorities; do not mix them in the same reservoir.

The target profile

Per gallon, full-strength fruiting solution:

Macro	Target	Micro	Target
Nitrogen (N)	116 ppm	Iron (Fe)	2.4 ppm
Phosphorus (P)	47 ppm	Boron (B)	1.2 ppm
Potassium (K)	188 ppm	Manganese (Mn)	1.2 ppm
Calcium (Ca)	113 ppm	Zinc (Zn)	0.3 ppm
Magnesium (Mg)	41 ppm	Copper (Cu)	0.3 ppm
Sulfur (S)	58 ppm	Molybdenum (Mo)	0.06 ppm

When to pick which

Pick Masterblend if…	Pick DIY if…
You want a turnkey system that works the first time	You enjoy the foraging process and want a local closed loop
You are running a 20+ bucket commercial-scale operation	You are running a small, attention-rich household garden
You need predictable, repeatable results between batches	You are willing to take occasional crop hits as the price of learning
Your nearest oak woodland is far away	You have oak woodland on the property
You don’t have time to brew teas	You have a bat colony nearby (Kartchner area, etc.)

For most growers the practical answer is: **start with Masterblend for the first year**, get comfortable with EC and pH meters, learn what a healthy plant looks like in hydroponics. **Add DIY substitutions selectively in year two** — start with oak-ash potassium and prickly-pear vinegar pH Down, both of which are forgiving. Move other inputs to local sources as you build confidence with each one.

RELATED GUIDES

For the full Masterblend program — per-crop feeding charts, stock-solution preparation, EC and pH daily protocol, deficiency diagnosis — see **Guide 3 Part 1**. **For the DIY locally-sourced recipes** — oak ash, bat guano tea, caliche calcium, epsomite magnesium, the 8-step DIY workflow — see **Guide 3 Part 2**. The two Parts share a single ppm target profile.

pH — short version

Why it matters · target ranges · adjustment recipes → Guide 3 Part 3

pH is the second-most-consequential daily variable after water hardness. Outside the right pH range, even a perfectly mixed nutrient solution becomes effectively unavailable to the plant — iron, manganese, and phosphorus all lock up at high pH, and calcium and magnesium become less available at low pH. The good news is that pH responds quickly and predictably to small doses of acid or base **once the bicarbonates have been neutralized first** (see §6).

Target ranges

Growing context	Target pH	Why
Hydroponic fruiting crops (peppers, tomatoes, cucumbers)	5.8 – 6.2	Iron and phosphorus most available; matches Masterblend chemistry
Hydroponic leafy greens and herbs	5.8 – 6.3	Slightly wider tolerance; lettuce slightly higher than peppers
Hydroponic brassicas	6.0 – 6.5	Brassicas tolerate (and prefer) slightly higher than fruiting crops
In-ground soil — most vegetables	6.5 – 7.0	Microbial activity and nutrient cycling optimal
In-ground soil — brassicas	6.5 – 7.5	Higher pH suppresses clubroot disease
In-ground soil — grapes	6.0 – 7.0	Lower than most vegetable crops; sulfur amendment usually needed
In-ground soil — Rocoto and other Capsicum	6.0 – 6.5	Sulfur amendment needed against native 7.5–8.5
In-ground soil — alliums	6.2 – 7.0	Moderately flexible

LOCAL TIP

Drift direction tells the story. In a healthy hydroponic reservoir on well-treated water, pH drifts slowly upward over a week as plants consume nitrate (which releases hydroxide). If yours is drifting upward fast, bicarbonates were not fully neutralized in the original treatment. If it drifts downward, ammonium-N is being absorbed faster than nitrate, or the reservoir has gone biologically off. **Watch the direction, not just the number.**

Acid and base options at a glance

pH Down	Best for	pH Up	Best for
Citric acid stock (1 g/mL)	Standard daily use; accurate dosing	Potassium hydroxide stock	Standard daily use; accurate dosing
Phosphoric acid (commercial)	Hardest well water; adds useful P at fruiting	Oak-ash lye (foraged)	Local source; adds useful K at fruiting
Prickly-pear vinegar (foraged)	Gentle daily nudges; unlimited local supply	Hydrated lime (foraged caliche)	Backup; adds Ca
White vinegar	Budget option	Baking soda	Emergency only — sodium accumulates

RELATED GUIDES

Full recipes for every acid and base — quantities, fermentation times, safety notes, dilution ratios, and the universal dosing protocol — live in **Guide 3 Part 3**.

Master Dragoons multi-crop calendar

The single-source-of-truth for "what is going on in the garden right now"

This is the synthesized calendar that ties the whole library together. Each crop guide has its own per-crop calendar, but in practice you do not garden one crop at a time — you garden five or six families at once and need to know what to do **this week** across all of them. The table below answers that question.

The calendar reflects east-side Dragoons at roughly 4,700 ft elevation, USDA zone 8a. Shift dates 1–2 weeks earlier at lower elevations (4,200–4,500 ft, valley floors) and 1–2 weeks later at higher elevations (5,000–5,500 ft). For exact per-crop dates, see the specific guide.

The master table

Month	Peppers (Guide 1)	Grapes (Guide 5)	Brassicas (Guide 6)	Alliums (Guide 6)	Greens (Guide 6)	Roots (Guide 6)
Jan	Start Rocoto + habanero seeds indoors under LED; apply sulfur to outdoor beds if pH > 7.5	Full dormancy; plan pruning strategy; soil samples; sulfur amendment	Harvest mature heads on warm days; protect during 25°F nights	Garlic dormant under mulch; harvest stored bulbs	Harvest spinach, mâche, arugula under row cover	Pull carrots, beets, parsnips as needed from in-ground storage
Feb	Start all *C. annuum* seeds indoors; clean reservoirs; order Masterblend stocks	Late-dormancy pruning (delay to dodge frost); compost around bases; frost cloth standby	Sow spring brassicas indoors (Feb 1–15)	Garlic still dormant; sow spring onions/leeks indoors (Jan 15–Feb 1)	Continue winter harvest; sow successions of arugula, spinach	Sow successions of radishes

Month	Peppers (Guide 1)	Grapes (Guide 5)	Brassicas (Guide 6)	Alliums (Guide 6)	Greens (Guide 6)	Roots (Guide 6)
Mar	Pot up seedlings; build/prep outdoor systems; set up monsoon catchment; last sulfur application	Bud break — **FROST WATCH**; begin drip irrigation; first N application; frost cloth ready	Transplant spring brassicas (Mar 1–15); direct-sow broccoli	Transplant spring onions/leeks (Mar 1–15); side-dress fall garlic with blood meal	Sow lettuce, radish, arugula successions outside	Sow carrots, beets (Feb 15–Mar 15)
Apr	Harden off; check soil temp 60°F+; transplant late April after last frost; install drip	Active shoot growth; shoot positioning on trellis; **frost still possible** — keep cloth	Spring brassicas race against May heat; harvest overwintered crops	Side-dress garlic with N (last app before bulbing); fall garlic scapes coming	Last spring sowings of greens; transition beds	Spring root crops continue
May	Final transplants out; mulch heavily; first fertilizer; watch late frost until mid-month	**Flowering — CRITICAL MONTH**; even irrigation; boron foliar before bloom; no copper during bloom	Cool-season crops bolt; bed turnover for peppers; spring brassicas wrap up	Garlic scapes harvest (cut for flavor + bulb size); pull spring onions	Bolt watch — pull and replace before going to seed	Pull spring carrots, beets as ready
Jun	Peak pre-monsoon heat; first flower buds; low-N switch; shade cloth on if > 95°F	Fruit set complete; canopy management; prep monsoon catchment; rising irrigation	Beds resting or sown to summer cover; first brussels sprouts indoor sowing late June	Garlic ready: bulb-formation complete; harvest hard-neck Jun 20–Jul 10	Few greens this month; basil takes over the bed	Beds resting; soil prep for fall sowings

Month	Peppers (Guide 1)	Grapes (Guide 5)	Brassicas (Guide 6)	Alliums (Guide 6)	Greens (Guide 6)	Roots (Guide 6)
Jul	**Monsoon begins**; reduce drip; stake heavy plants; Rocoto flowering heavily; aphid watch	Monsoon arrives mid-month; reduce drip; **powdery mildew watch**; sulfur/copper preventive	Sow fall brassicas indoors (Jul 5–Aug 5 — broccoli, cabbage, brussels)	Soft-neck and shallot harvest Jul 1–20; cure for 3 weeks	Sow leek transplants indoors (Jul 1–20) for fall transplant	Soil prep finalized for Aug-Sep direct sowings
Aug	Peak production *C. annuum*; harvest jalapeños, serranos, cayenne; habaneros start; Rocoto approaching	**Veraison** (color change); reduce to deficit irrigation; weekly Brix; leaf-strip east side; white varieties harvest late Aug	Sow more fall brassicas indoors; transplant first brassicas out (Aug 25+); sow fall kale	Transplant overwintering onions (Aug 25–Sep 15)	Late August: prep for September direct-sow window	Late August: prep beds; soil cooling toward 65°F
Sep	● **BEST MONTH** — peak harvest all varieties; cool nights return; Rocoto ripe	● **PRIMARY HARVEST MONTH** — reds ready; pick by Brix; harvest early morning; post-harvest fish emulsion	Transplant brassicas all month; direct-sow broccoli through Sep 25; sow cabbage; sow kale Sep 1–Oct 15	Transplant overwintering onions/leeks early Sep; sow scallions; bunching onions	**Direct-sow window opens**: spinach, lettuce, arugula, mustards Sep 1+	Direct-sow carrots, beets, turnips, parsnips, radishes — primary fall sowing window

Month	Peppers (Guide 1)	Grapes (Guide 5)	Brassicas (Guide 6)	Alliums (Guide 6)	Greens (Guide 6)	Roots (Guide 6)
Oct	Continue harvest; first frost late Oct possible; pot up Rocoto and over-wintering plants; pull last vines	Continue irrigation until leaf fall; compost around vines; plant cover crop; last sulfur app	Harvest first fall heads (late Oct); transplant last brassicas early Oct only	**Plant garlic** (Oct 15–Nov 15); plant shallot sets; plant overwin-tering leek transplants	Continue lettuce, spinach, arugula sowings under row cover	Continue carrot, beet, radish sowings through Oct 15; sow mâche
Nov	Frost ends outdoors; bring over-wintering plants in under LED; dry remaining harvest; compost beds	Leaf fall; irrigation taper to monthly deep soak; mow/terminate cover crop; soil test; return grape marc to compost	Heavy fall harvest — broccoli, cauliflower, cabbage, kale; row cover ready	Garlic and shallots es-tablishing in-ground; last fall planting	Continuous harvest under row cover; mâche thriving	First fall carrot harvest; beets, turnips ready; storage in-ground
Dec	Overwintered plants under LED (16 hr); EC 1.0–1.4; minimal feeding; plan next season	Full dormancy; final amend-ments; review season notes; plan variety additions	Harvest mature heads on warm days; brussels sprouts at peak after frost	Garlic dormant; mulch in place; harvest stored bulbs	Spinach, mâche, kale continuous harvest; cilantro thriving	Pull as needed; parsnips developing sweetness with each freeze

Per-month detail callouts

The table above is the navigation surface. The notes below are the at-a-glance reasons behind each row — the *why* behind the *what*.

January — the planning month. The garden is mostly dormant outdoors. The most consequential January action is starting Rocoto and other long-season peppers under LED indoors; they need 10–14 weeks before transplant and germination is slow (14–

28 days). January is also the right month for elemental sulfur on alkaline soil-track beds — sulfur takes weeks of bacterial action to lower pH and needs to be in place before spring planting.

February — start everything that wants a long indoor head start. All *C. annuum* peppers and tomatoes, spring brassicas, spring onions and leeks. Outdoors, the garden is mostly waiting; this is when reservoir cleaning, system inspection, and material ordering happen.

March — bud break and the start of the active season. Grape bud break is the single highest-stakes event of the spring — once buds break, frost damage becomes irreversible. Spring brassicas and onions go out under row cover. Last chance for sulfur amendment before warm-season planting.

April — the bridge month. Hardening off, transplanting peppers, racing spring brassicas. Frost is still possible into mid-April at higher elevations, so frost cloth stays accessible. Drip irrigation goes in.

May — the flowering and fruiting setup month. Grapes flower (the single most weather-dependent event of the grape year). Peppers and tomatoes establish in beds and buckets. Spring brassicas bolt by Memorial Day. Garlic finishes its top-growth phase and begins moving energy into bulbs.

June — the pre-monsoon stress month. The hardest month for the garden. Heat hits 95°F+ but the monsoon hasn't started. Reservoir-temperature management peaks. Garlic harvest begins. Shade cloth goes on. Cool-season beds are empty or in summer cover crop.

July — the monsoon arrives. Mid-July through mid-September the climate shifts: 50–70% humidity, daily afternoon storms, soft rainwater. Soft-neck garlic and shallots harvest and cure. Fall brassica seeds start indoors. Powdery mildew watch begins on grapes.

August — peak summer production + fall setup. *C. annuum* peppers hit peak harvest. Grape veraison. First fall brassicas transplant out the last week. Overwintering onions go in late August through mid-September. Beds get prepped for the September direct-sow blitz.

September — the Dragoons' best month. Peak grape harvest, peak pepper harvest, the cool-season direct-sow window opens. The single most important month in the calendar. Plan around it.

October — turnover. First frost arrives late in the month. Garlic goes in mid-October. Last brassica transplants early in the month. Peppers wind down; Rocotos come indoors. Cover crops sown between grape rows.

November — fall harvest and overwintering setup. Frost ends the warm-season garden but the cool-season garden is at peak. Brassicas, kale, brussels sprouts, root crops, greens — all harvesting heavily. Indoor overwintering peppers settle into LED routines.

December — dormancy with continuous cool-season harvest. Grapes fully dormant. Peppers indoors. Outside, the cool-season garden under row cover keeps producing greens, brassicas, and roots through the deepest weeks of winter.

Crop selection beyond the named guides

Warm-season fruiting crops not covered in their own chapter

The six named guides cover most of what you will actually grow at the Dragoons — peppers, grapes, brassicas, alliums, greens, and roots. The crops below are the secondary warm-season fruiting and herb crops that share growing techniques with the named guides without needing a guide of their own. Treat them as Guide 1 (peppers) or Guide 2 (Bato system) techniques applied to a slightly different crop.

Tomatoes

A natural pairing with peppers — same warm season, same Bato bucket system, similar nutrient program scaled to higher EC. Indeterminate slicing tomatoes work in 18–24” Bato spacing on an overhead-twine trellis (see Guide 2 §7 for the trellis spec). Heat-tolerant cherry varieties (Sungold, Solar Fire, Heatmaster) set fruit even at 95°F; classic large-fruited beefsteaks struggle in June heat. Plant for fall harvest where possible — September tomatoes from a June transplant are the best you will grow.

- **Dragoons fit:** Excellent in spring and fall; needs shade and reservoir cooling in June.
- **Cross-ref:** Guide 2 entire (system); Guide 3 Part 1 §1.4.1 (full tomato nutrient program); Guide 1 §6 (hydroponic operating principles transfer directly).

Cucumbers

Possibly the perfect monsoon crop — humidity and warm rain are exactly what cucumbers want. Plant Marketmore, Diva, or Suyo Long in mid-May for July harvest, or start a second crop in late July for September. Trellis vertically; the same overhead-twine system that works for tomatoes works for cucumbers.

- **Dragoons fit:** Loves the monsoon; struggles in dry June; pre-monsoon irrigation is critical.
- **Cross-ref:** Guide 2 (Bato build works directly); Guide 3 Part 1 (use tomato program with slightly lower EC).

Basil and warm-season herbs

Basil is the single best summer/monsoon herb here — it loves the heat, loves the humidity, and tolerates high alkaline water with minimal fuss. Genovese, Thai, and Holy basil all thrive. Thai basil and Mexican oregano are particularly well-adapted. Cilantro is the opposite — strictly a cool-season crop in the Dragoons, bolting fast above 80°F.

- **Dragoons fit:** Basil is the summer easy-mode crop; cilantro and dill are cool-season.

- **Cross-ref:** Guide 3 Part 1 §1.4.3 (leafy/herb nutrient program); Guide 3 Part 1 §1.8 (basil-specific optimization).

Eggplant

Heat-loving and Dragoon-friendly. Long-season — start indoors in February alongside peppers. Italian and Japanese types both work; Japanese (Ichiban, Millionaire) sets fruit more reliably in the heat. Bato buckets work well with same setup as peppers.

- **Dragoons fit:** Good — wants similar conditions to peppers but a longer warm season.
- **Cross-ref:** Guide 1 §6 (hydroponic) and §5 (soil) techniques apply directly with no major changes.

Summer squash and zucchini

Productive in raised beds; not commonly hydroponic. Vines need wide spacing (3 ft) and lots of water during monsoon. Vine-borer-resistant varieties (Tatume, Cucuzza, tromboncino) outperform classic zucchini at our elevation. Direct-sow after last frost (mid-April at lower elevations) for July harvest.

- **Dragoons fit:** Easy in raised beds; powdery mildew watch in monsoon.
- **Cross-ref:** Guide 6 §1 raised-bed soil chemistry transfers directly.

Beans (bush and pole)

Direct-sow after last frost; pole beans through June; bush varieties through July. Tepary beans and other Sonoran heritage beans are exceptionally well-adapted to the dry season — they outperform standard green beans in low-water conditions. Heritage Apache and Tohono O'odham varieties are available from Native Seed/SEARCH in Tucson.

- **Dragoons fit:** Tepary beans are essentially perfect; standard green beans are good.
- **Cross-ref:** Soil-bed prep follows Guide 6 §1 principles; no nutrient guide needed (legumes fix their own N).

Sweet corn

Possible but never trivial in the Dragoons. Needs heavy water during silking (late June to mid-July, right at the pre-monsoon dry peak) and competes with the warm-season fruiting crops for bed space. If you grow it, plant in blocks (not rows) for pollination and time the silking to overlap with monsoon arrival rather than precede it.

- **Dragoons fit:** Workable but a stretch — most growers skip it.
- **Cross-ref:** Soil prep follows Guide 6 §1 principles.

Native and heritage staples

The Dragoons sit at the intersection of three native agricultural traditions — Apache, Tohono O'odham, and Pueblo. Heritage staples worth growing include Chiltepin (Guide 1 §2), tepary beans (above), amaranth (heat-tolerant grain green), and devil's claw. These crops were bred for exactly the conditions you are growing in — they tend to outperform supermarket varieties at our elevation with less water and zero pampering.

> **LOCAL TIP**
>
> Source heritage Southwest varieties from Native Seeds/SEARCH (Tucson) or local seed swaps. They are routinely better-adapted to Cochise County than commercial seed-catalog varieties, often by a wide margin.

Troubleshooting decision tree

When something is wrong — where to look

This is the navigation table for problems. Each row maps a symptom to the guide and section that contains the actual diagnosis and fix. The principle: this primer routes you to the right specialist, but the specialist is where the answer lives.

By visible symptom

Symptom	Likely cause	Where to read
Yellow leaves at the **bottom** of pepper plants	Nitrogen deficiency (common in fruiting peppers)	**Guide 1 §6.5** deficiency reference; **Guide 3 Part 1 §1.7** for the full deficiency table
Yellow leaves with **green veins** on new growth	Iron deficiency from high pH or bicarbonates	**Guide 2 §4** (water treatment); **Guide 3 Part 1 §1.7** (Fe deficiency)
White crust on reservoir walls / cloudy reservoir within 24 hrs	Bicarbonates not fully neutralized; or wrong mix order	**Guide 2 §4** — review Steps 3 and 5 carefully
pH climbs back overnight after adjustment	Step 3 of bicarbonate scrub truncated — wait longer	**Guide 2 §4** Step 3
Brown leaf edges / tip burn on lettuce or peppers	Salt burn (high EC) or calcium deficiency	**Guide 3 Part 1 §1.7**; check EC meter first
Wilting despite wet roots	Root rot from warm reservoir (>75°F) or pythium	**Guide 1 §6.2** (reservoir temperature); **Guide 2 §6** (warning callout on disease)
Blossom drop on peppers (or tomatoes)	Heat stress, humidity stress, or boron deficiency	**Guide 1 §5.3** (soil) and **§6.6** (hydroponic)
Blossom end rot on peppers / tomatoes	Inconsistent water (soil); calcium availability	**Guide 1 §5.4** “calcium management”
Broccoli **buttoning** (tiny premature head)	Plants stressed by heat or transplanted too small	**Guide 6 §2.5** (broccoli) — review timing
Brassica heads **splitting**	Heat stress; uneven water; over-mature	**Guide 6 §2.5** — harvest sooner; mulch heavily
Aphid colonies in monsoon	High humidity + new growth	**Guide 1 §8** (peppers); **Guide 6 §6** (brassicas/cool-season)

Symptom	Likely cause	Where to read
Powdery white coating on grape or cucurbit leaves	Powdery mildew — monsoon humidity	**Guide 5 §10** (grape); **Guide 6 §6** (squash family) — sulfur or apple-cider vinegar spray
White cabbage butterflies all over brassicas	*Pieris rapae* — primary fall brassica pest	**Guide 6 §6.2** — row cover, *Bt*, hand-pick
Garlic bulbs small / single-clove	Planted too late or insufficient cold	**Guide 6 §3** — review October planting date; this guide §9
Carrots forking / stubby	Caliche or rocks in bed; insufficient depth prep	**Guide 6 §5.3** — soil-prep is the make-or-break factor
Brussels sprouts not forming	Insufficient frost exposure or planted too late	**Guide 6 §2** — need August transplant for full winter season
Rocoto not setting fruit in summer	Daytime heat above 85°F; needs cool nights	**Guide 1 §3** (Rocoto special section)
Grape vine no fruit second year	Normal — vines fruit year 3	**Guide 5 §12** (year 2+ dormancy)
Slow growth in summer despite healthy color	Heat stress above 95°F	This guide §4 — review climate; install shade cloth
Oak ash lye is not bringing pH up enough	Ash too old, leach too dilute, or buffering against bicarbonates	**Guide 3 Part 3 §3.3.1** — review concentration
Prickly-pear vinegar is not bringing pH down enough	Fermentation incomplete or batch too dilute	**Guide 3 Part 3 §3.2.1** — check Brix/pH of batch
Drip emitters clogging on Bato system	Calcium nitrate scale	**Guide 2 §9** — vinegar soak; weekly maintenance
Algae bloom in reservoir	Light reaching solution	**Guide 2 §9** — opaque tub + lid
Solar pump not running scheduled cycle	Battery undercharged or panel obstructed	**Guide 2 §9** — clean panel, check angle
Bolting greens in spring	Daylength + heat	**Guide 6 §4** — succession plan; choose slow-bolt varieties

By guide if you do not know where to start

If your problem is…	Read first
Anything about pepper cultivation (any system)	Guide 1
Anything about a Bato bucket build or operations	Guide 2
Anything about nutrient mixing or deficiency	Guide 3 Parts 1 (Masterblend) or 2 (DIY)
Anything about pH or water-treatment chemistry	Guide 3 Part 3 (recipes), Guide 2 §4 (procedure)
Anything about LED, indoor, VPD, or CO_2	Guide 3 Part 4
Anything about a grape vine	Guide 5
Anything about cool-season crops (Sep–May)	Guide 6
Anything about a crop not in the named guides	This guide §10

The 80/20 rule for Dragoon problems: roughly 80% of problems trace back to one of three root causes — untreated bicarbonates (Guide 2 §4), reservoir temperature too high (Guide 1 §6.2 or Guide 2 §6), or the wrong nutrient EC for the crop and stage (Guide 3 Part 1 §1.4). Check those three first before chasing more exotic explanations.

Closing

The Dragoon Mountains are an exceptional place to grow food. The monsoon gives you soft water, the elevation gives you cool summers and a true winter, the local geology gives you nearly every element you need, and the long, clear fall gives you the single best growing window in the American Southwest. Match crops to seasons, treat the water, watch the meters, and pay attention to which guide answers which question.

This primer is deliberately thin. It is the map, not the terrain. When you want to build a system, brew a nutrient, ferment an acid, prune a vine, or understand why broccoli buttons when transplanted late, step sideways into the specialist guide that covers it. The library is designed that way — shallow here, deep where it matters.

Plant for fall. Collect the monsoon. Treat the water. The rest follows.